

Nirbhay Singh

Senior Radar & FPGA Systems Engineer @ Aero space and Defense



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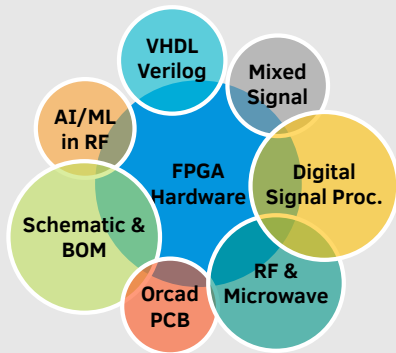
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Technical Skills

Overview



Programming

Embedded C • JESD204 • SerDes

Python • Deep Learning • ML for Radar

MATLAB • Simulink • DSP

VHDL • FPGA • RFSoc

Education

M.Tech., Communication & Signal Processing

Indian Institute of Technology (IIT) Madras
CGPA: 9.32 / 10

B.Tech., Electronics & Communication Engineering

Bundelkhand Institute of Engineering & Technology
80% Graduated with Honours

About Me

Senior Mixed-Signal Hardware and FPGA Engineer with 10+ years of experience in designing, developing, and leading high-performance defence and aerospace electronic systems. Expertise in mixed-signal hardware architecture, RF systems, high-speed digital design, FPGA development.

Experienced in designing direct RF sampling systems, high-speed ADC/DAC interfaces (up to 12 GSPS), JESD204B/C multi-board synchronization, clock distribution, and embedded hardware platforms. Strong background in schematic design, PCB layout reviews, component selection, signal integrity (SI), power integrity (PI), power management, and hardware debugging.

Proven ability to lead multidisciplinary teams, define hardware architecture, collaborate with firmware and software engineers, and deliver reliable, embedded hardware solutions.

Experience -- 10+ Years

Nov 2023 – Present **Deputy Manager – Radar Systems R&D** Bharat Electronics Limited (BEL)

- Led the hardware architecture and development of complex mixed-signal electronic systems for defence applications.
- Architected and validated **RFSoc-based multi-channel digital beamforming (DBF) systems** supporting up to **136 channels**
- Designed high-speed mixed-signal boards integrating RF ADCs, DACs, FPGA, clock generators, power supplies, and RF front-end circuitry
- Developed complete hardware schematics, performed PCB design reviews, and optimized layouts for high-speed signal integrity and power integrity.
- Developed **multi-board deterministic latency synchronization** solutions critical for large AESA arrays
- Collaborated closely with firmware teams for SPI, I2C, UART, GPIO, and FPGA interface development.
- Coordinated cross-functional teams including hardware, firmware, FPGA, RF, mechanical, and manufacturing engineers.

2017 – 2022 **Senior Hardware& Digital Systems Engineer** Bharat Electronics Limited (BEL)

- Designed mixed-signal receiver and transmitter hardware for phased-array radar systems.
- Developed FPGA firmware interfacing with high-speed ADCs, DACs, DDR memory, Ethernet, and embedded peripherals.
- Performed RF system analysis including gain budgeting, noise figure, dynamic range, clock jitter, and phase noise optimization.
- Conducted schematic reviews, PCB verification, hardware debugging, and system integration.
- Designed communication interfaces using SPI, I2C, UART, JESD204, Ethernet, and GPIO.
- Supported EMC/EMI compliance, reliability testing, and field validation

2015 – 2016 **Mixed Signal Processing & RF Design** Bharat Electronics Limited (BEL)

- Designed analog and digital hardware modules for RF transceiver systems.
- Performed schematic capture, component selection, and mixed-signal circuit analysis.
- Conducted signal integrity, power integrity, and power budgeting analysis.
- Supported PCB bring-up, laboratory validation, and system debugging.

Technical Skills

Hardware Design

- Mixed-Signal Hardware Architecture
- High-Speed Digital Design
- Analog and RF Circuit Design
- PCB Design and Schematic Capture
- Component Selection and BOM Management.
- Signal Integrity (SI), Power Integrity (PI)
- Power Supply Design
- Hardware Debugging
- Prototype Development
- Manufacturing Support

Embedded Systems

- FPGA (VHDL, Verilog)
- Embedded Hardware Interfaces
- SPI, I2C, UART, GPIO
- Ethernet(1G, 10G,40G)
- JESD204B/C.

Mixed Signal

- RF Receiver and Exciter Design
- Ultra-Low Phase Noise Systems.
- RF ADC/DAC up to 12 GSPS
- Clock Tree Design
- Synchronization
- Timing Distribution

Simulation and Tools

- MATLAB/Simulink, ANSYS HFSS
- Python, Cadence/Orcad
- Autocad, Actcad

Leadership

- Hardware Architecture
- Technical Team Leadership
- Design Reviews
- Cross-functional Collaboration
- Product Planning and Validation

Research

- 2022 – 2024 **Postgraduate Research – Radar AI** Indian Institute of Technology Madras
Thesis: Target Detection and Classification Using Deep Learning for Surveillance Radar
- Deep learning models for radar target detection and classification
 - Analysis of phase noise and micro-Doppler effects on ML-based radar systems
 - Tools: MATLAB, Python, Deep Learning Toolboxes

Publications

- **Nirbhay Kumar Singh**, “Advanced Deep Learning Technique for Digital Predistortion to Improve Power Amplifier Linearity”. IEEE MAPCON2024
- **Nirbhay Kumar Singh**, Malwinder Singh, Shaik Abdul Subhan, Seema, “Phase Noise Effects on Micro-Doppler Extraction for Radar Target Classification Using Machine Learning”. IEEE APMC2024, Bali, Indonesia
- **Nirbhay Kumar Singh**, “Multi-board Synchronization of n-channel Direct RF Digital Transceiver for Active Phased Array Radar using JESD204 Interface”. IEEE I2CT · Apr 6, 2024
- **Nirbhay Kumar Singh**, Vikas Kumar, T Venakataamuni, “Ultra-Low Phase Noise, Digital Transceiver for High Range Resolution Pulsed Agile Radar”. 8th IEEE International Radio Conference, Mauritius · May 1, 2023
- **Nirbhay Kumar Singh**, T Venakataamuni, “A Wide-band SWaP System for VSWR and Power Measurement with Temperature Compensation Technique”. IEEE MAPCON2022 · Dec 14, 2022
- **Nirbhay Kumar Singh**, Vikas Kumar, Fouziya c, T Venakataamuni, A Programmable, Multimode Operational 3U-VPX Based Digital Transceiver & Processing Module For CIT-MKXIIA IFF
- **Nirbhay Kumar Singh**, T Venakataamuni, “A method to Enhance Power Amplifier Linearity using Time delay estimation & correction Technique”, IEEE, 2022 International Conference for Advancement in Technology (ICONAT) Goa, India. Jan 21-22, 2022
- Vikas Kumar, **Nirbhay Kumar Singh**, T Venakataamuni, “HIGH-RESOLUTION FMCW GENERATION FOR MICRO DOPPLER IN DRONE DETECTION RADAR”, International Radar Symposium India(IRSI) 2019 9. **Nirbhay Kumar Singh**, Vikas Kumar, Fouziya c, T Venakataamuni, “Performance Analysis of High-Resolution Radar Waveforms for Naval Tracking Radar”. EWCI 2018
- **Nirbhay Kumar Singh**, Fouziya C, Venkataamuni, “Radar Target Emulator For Naval Tracking Radar Using FPGA”, URSI-RCRS 2017

Patents

- A method and system of synchronization of an n-channel direct RF digital transceiver with high-speed serial JESD204 protocol. Patent no: 202341083757
- A Method to generate a wideband exciter signal having multi-pulse-coded waveforms with test-target simulation in a radar receiver. Patent no: 202341004980
- Wide-band SWAP System and Method for VSWR and Power Measurement using Digital Signal Processing with Temperature Compensation. Patent no: 202241070066
- Compact and Programmable Digital Transceiver for Software-defined Radar. Patent no: 202041013933
- System and Method to Convert Non-Coherent FMCW Radar to Coherent Radar. Patent no: 202141009908
- A SYSTEM AND METHOD FOR REALIZING ULTRA-LOW PHASE NOISE, FAST SETTLING, COMPACT PHASE COHERENT AND ENERGY EFFICIENT DIGITAL SOURCE GENERATOR. Patent no: 202341030479

Awards

- Nominated for the IETE Young Scientist Award, one of India's highest honors recognizing outstanding contributions by young scientists..
- R&D Excellence award 2020-23(Excellent Technology project): MKXIIA-CIT.
- R&D Excellence award 2020-23(Innovation category): RFSoc based Integrated Processor Module.
- R&D Excellence award 2019-20(Innovation category): Programmable RF and SoC based processing module for IFF MKXIIA - CIT.
- R&D Excellence award 2016-17(Innovation category): DIRECT SAMPLING DIGITAL RECEIVER for IFF.
- Best Technical Poster Awards (Defence AI, Unmanned Systems)

Additional Information

- Languages: English (Professional), Hindi (Native)
- Open for relocation; Skilled Worker Visa sponsorship required
- 10+ years experience in Defence & Aerospace R&D